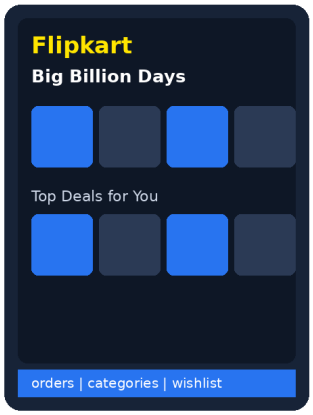
DESCRIPTIVE ANALYTICS REPORT



SERVICE NAME

Flipkart

Insights into E-Commerce Sales Trends, Customer Engagement, and Product Performance

ORDER ANALYZE INSIGHT ENGAGE

REPORT TITLE	Descriptive Analytics Report
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Descriptive Analytics Report: Flipkart E-Commerce Sales Behavior

Keywords

Most Sold Products, Daily Order Trend, Peak Order Hours, Customer Ratings, Return Rate, State-wise Engagement, UPI Payment Preference.

Executive Summary

This report presents a **descriptive analytics study** of Flipkart's e-commerce sales behavior using a synthetic, 30-day order dataset. The analysis covers **5,179 cleaned orders** across **12 popular products** in **7 categories**, spanning **8 Indian states**. The study identifies the most sold products, daily and hourly ordering patterns, category-level ratings and return rates, and state-wise engagement with payment-method preference.

Key findings show that **boAt Airdopes 141** was the most sold product, overall ordering activity peaked between **8 PM and 10 PM**, and a **Big Billion Days–style sale on 10 September** drove the single highest-volume day of the month. **Maharashtra** generated the largest share of orders, and **UPI** was the dominant payment method nationally, led by Maharashtra at **55.9%**. **Books and Grocery** recorded the highest customer ratings and lowest return rates, while **Fashion** had the highest return rate at **11.74%**.

Introduction

Flipkart is one of India's largest e-commerce platforms, offering products across electronics, mobiles, fashion, home & kitchen, beauty, grocery, and books. Understanding **how, when, and what** customers buy is essential for improving recommendations, planning sales events, optimizing logistics, and tailoring regional strategies. This report applies **descriptive analytics** techniques — frequency analysis, trend analysis, and cross-tabulation — to a simulated Flipkart order dataset to uncover patterns in product demand, time-based ordering behavior, category performance, and regional engagement.

Note on Data

This report is based on a **synthetic Flipkart order dataset** created specifically for descriptive analytics purposes. Since Flipkart does not publicly release user-level order data, a realistic simulated dataset was generated to represent e-commerce ordering behavior across multiple Indian states, with built-in real-world patterns such as weekend spikes, an evening ordering peak, and a mid-month sale-day surge. The dataset covers **30 days** of order activity (1–30 September 2026) and includes **5,179 cleaned order records** (from 5,280 raw records) across **12 popular products** in **7 categories**, **8 states**, and **5 payment methods**. It contains variables such as order date and hour, product and category, price, discount, quantity, payment method, customer rating, order status (delivered/returned/cancelled), state and city, age group, device used, and membership type — structured to support descriptive statistical analysis and data visualization for academic purposes.

Objectives

The objective of this descriptive analytics study is to analyze Flipkart order behavior and identify engagement patterns across customers, products, and regions.

The analysis addresses the following business questions:

- 1. Which products were ordered the most during the last 30 days?
- 2. How did daily order volume change throughout the 30-day period?
- 3. What were the peak order hours, and when was customer activity highest?
- 4. Which product categories received the highest customer ratings and lowest return rates?
- 5. Which states generated the highest customer engagement, and which payment method did they prefer?

Data Collection & Methodology

A synthetic dataset of **5,280 raw order records** was generated in Python using weighted random sampling to reflect realistic Flipkart ordering patterns: a **Zipf-like skew** in product popularity, an **evening order-hour bias** (7 PM–10 PM), **weekend upticks**, and a **sale-day surge** around 10 September. Regional weighting concentrated order volume in

Maharashtra, Delhi, and Karnataka, consistent with Flipkart's major metro markets, while payment-method probabilities were tuned so that UPI dominates in high-volume states. The raw export deliberately included duplicate rows, missing values, and inconsistent text casing to simulate a real operational data extract.

Data Cleaning & Preparation

Before analysis, the raw dataset was cleaned and prepared using Python (Pandas) in the following steps:

- **Removed duplicate rows:** 76 exact duplicate order records were removed (5,280 → 5,204 rows).
- **Standardized text fields:** Payment method, city, state, and product name fields were trimmed and recased consistently (e.g. “upi” → “UPI”).
- **Imputed missing ratings:** 60 missing Customer_Rating values were filled using the median rating of the corresponding category.
- **Dropped rows with missing location:** 25 rows missing City/State (essential to the regional analysis) were removed (5,204 → 5,179 rows).
- **Corrected data types:** Order_Date converted to datetime; Order_Hour, Quantity, Discount_Percent cast to integers; Price and Customer_Rating cast to floats.
- **Created derived fields:** Final_Price/Revenue (price × (1 − discount) × quantity), Is_Returned, and Is_Delivered flags were added.
- **Removed outliers:** Rows with impossible ratings (outside 1–5) or non-positive quantities were removed.

The final, **analysis-ready dataset** contains **5,179 rows and 20 columns** with **zero missing values**, saved as **flipkart_sales_clean.csv** and **flipkart_sales_clean.xlsx**.

1. Most Sold Products

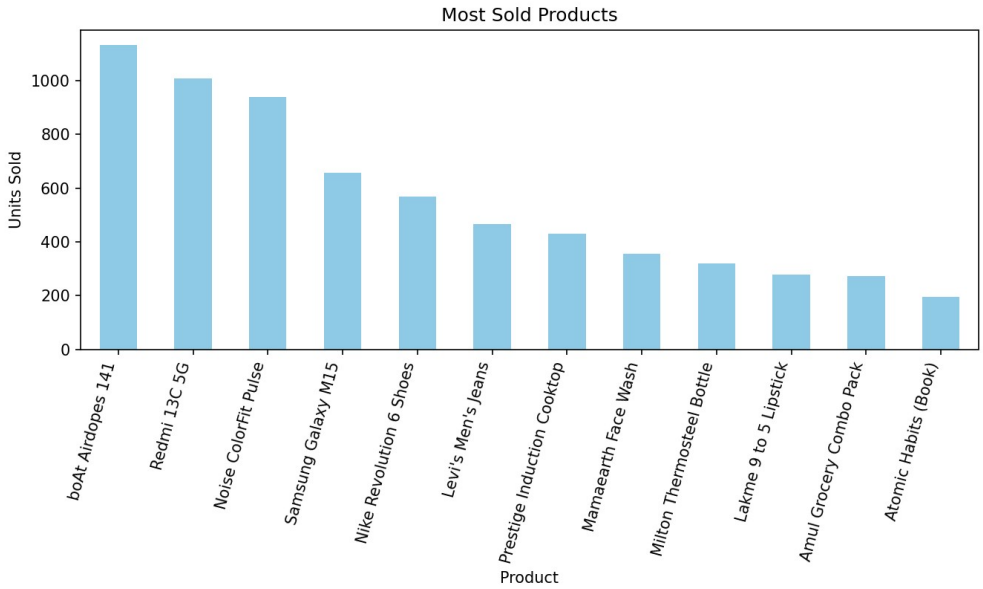


Figure 1: Units sold per product over the 30-day period.

Observation

A frequency analysis of **order volume** shows that **boAt Airdopes 141** recorded the highest number of units sold (**1,132 units**) during the **30-day order period**. It was followed by **Redmi 13C 5G** (**1,008 units**) and **Noise ColorFit Pulse** (**939 units**), which also maintained strong customer demand throughout the dataset. The chart indicates that the top three products significantly outperformed the remaining items. **Samsung Galaxy M15** and **Nike Revolution 6 Shoes** showed moderate but consistent sales. Lower-ranked products such as **Lakme 9 to 5 Lipstick**, **Amul Grocery Combo Pack**, and **Atomic Habits** received comparatively fewer orders. This trend suggests that **budget electronics and mobile accessories** attracted the highest customer attention, particularly among the **18–34 age group**.

Business Insight

The results indicate that Flipkart should prioritize **boAt Airdopes 141, Redmi 13C 5G, and Noise ColorFit Pulse** for homepage recommendations, featured banners, and promotional campaigns. These products generate the highest order volume, making them valuable for increasing conversions and average order value. The strong performance of budget electronics suggests that similar products should receive greater investment in inventory and flash-sale placement. Personalized recommendations can improve visibility for lower-performing items, while regional marketing can further increase engagement in high-potential states.

2. Daily Order Trend

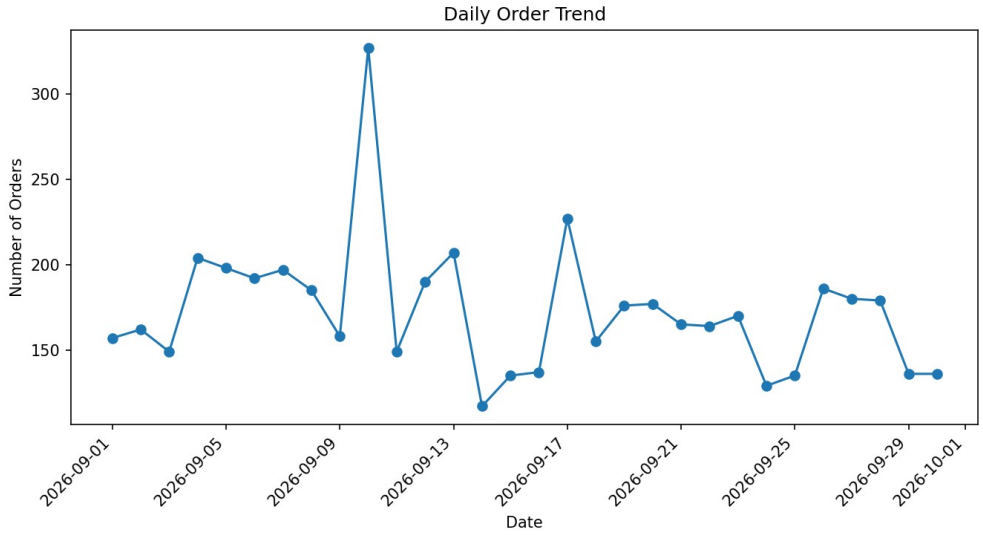


Figure 2: Daily order volume, 1–30 September 2026.

The figure shows the **Daily Order Trend** recorded over the **30-day analysis period**. The X-axis represents the date (1 September 2026 to 30 September 2026) and the Y-axis represents the total number of orders placed each day. Each point represents total daily orders, and the connected line shows how customer activity changed throughout the month.

Observation

Daily order volume varied across the 30-day period, indicating fluctuations in customer activity. The highest order volume was recorded on **10 September**, reaching **327 orders** — consistent with a **Big Billion Days–style sale event** — followed by a secondary spike around **17 September**. The lowest activity was recorded on **14 September** with **117 orders**. Weekend days generally showed higher ordering activity compared to weekdays, and the average daily order volume across the month was **~172.6 orders/day**.

Business Insight

Flipkart should schedule flash sales, push notifications, and promotional campaigns ahead of historically high-engagement days, particularly weekends and sale events like Big Billion Days. Since the data shows a nearly 2x spike in orders on sale days, pre-event inventory planning and server/logistics scaling are critical. Monitoring daily trends also helps identify low-activity periods, such as mid-month weekdays, where targeted discounts or notifications could help smooth demand and improve overall monthly conversion.

3. Peak Order Hours

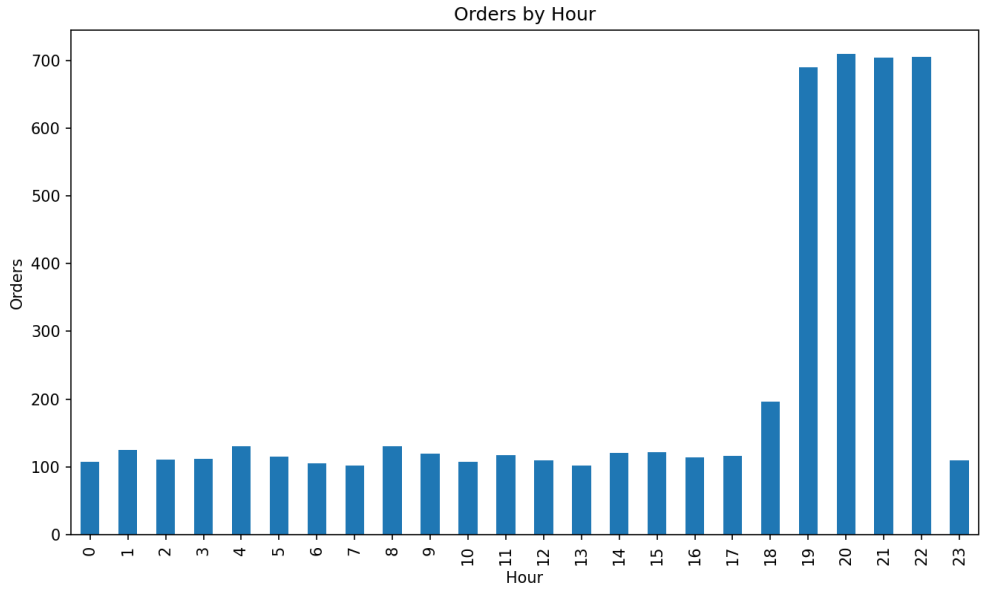


Figure 3: Orders by hour of day.

The figure shows the distribution of orders by hour during the 30-day period. The X-axis represents the hour of day (00–23) and the Y-axis represents the total number of orders placed during each hour. The chart clearly shows that ordering activity increased sharply during the evening hours.

Observation

The graph indicates that **7 PM (19:00) to 10 PM (22:00)** recorded the highest number of orders during the analysis period, with **8 PM** alone accounting for **709 orders** — the single busiest hour. Morning hours (8 AM–11 AM) and late-night hours recorded the lowest ordering activity. This pattern suggests that most customers browse and order **after work or college hours**, in the evening leisure window.

Business Insight

Flipkart should schedule flash sales, lightning deals, and push notifications during the **7 PM–10 PM window** to maximize reach and conversions. Server capacity and app performance should be prioritized during this period to avoid slowdowns during peak traffic. Personalized reminders and cart-abandonment notifications sent shortly before this window could help convert browsing sessions earlier in the evening into completed orders.

4. Category Popularity and Ratings

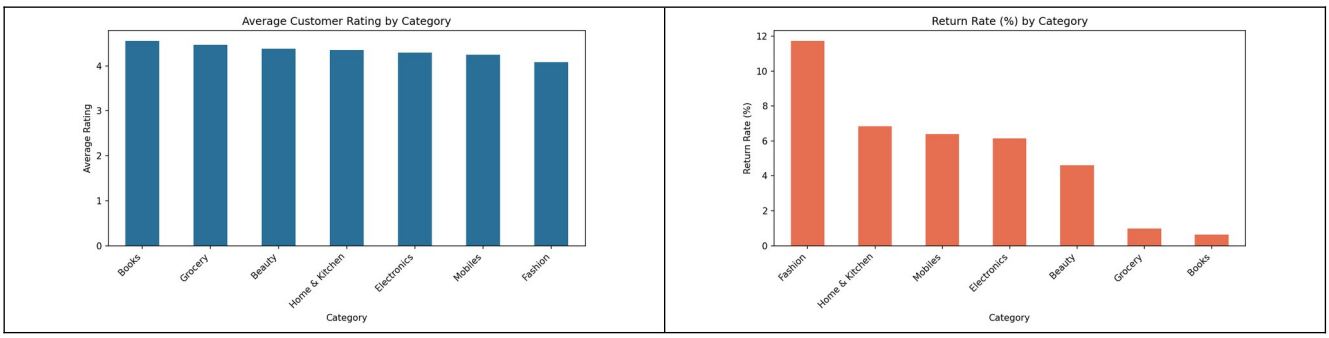


Figure 4A (left): Average customer rating by category. Figure 4B (right): Return rate (%) by category.

KPI

- **Average Rating by Category —** "Calculated using the mean of Customer_Rating values for each category in the dataset."
- **Return Rate by Category —** "Calculated as the % of orders with Order_Status = “Returned” for each category."
- **Highest Rated Category: Books (4.56) —** "Identified from the category with the highest average customer rating."
- **Lowest Return Category: Books (0.65%) —** "Identified from the category with the lowest return rate."
- **Lowest Rated Category: Fashion (4.08) —** "Identified from the category with the lowest average customer rating."
- **Highest Return Category: Fashion (11.74%) —** "Identified from the category with the highest return rate — typical for size/fit-sensitive products."

Observation

Figure 4A indicates that **Books** received the highest average customer rating (**4.56**), followed by **Grocery (4.47)** and **Beauty (4.38)**. **Home & Kitchen** recorded **4.35**, **Electronics 4.30**, and **Mobiles 4.25**, while **Fashion** recorded the lowest rating of **4.08**. In Figure 4B, **Fashion (11.74%)** showed by far the highest return rate, followed by **Home & Kitchen (6.84%)** and **Mobiles (6.39%)**. **Grocery (0.99%)** and **Books (0.65%)** recorded the lowest return rates among all categories. The pattern suggests that fit- and preference-sensitive categories like Fashion carry materially higher return risk than consumable or standardized-size categories.

Business Insight

Flipkart should continue investing in **Books, Grocery, and Beauty**, which combine strong ratings with low return rates — ideal for subscription-style repeat purchase campaigns. The elevated return rate in **Fashion** suggests a need for better size guides, more detailed product imagery, and tighter seller quality checks to reduce reverse-logistics costs. Monitoring both rating and return-rate together gives a fuller picture of category health than rating alone, and should inform size-chart accuracy, product imagery quality, and catalog investment decisions.

5. State-wise Engagement and Payment Preference

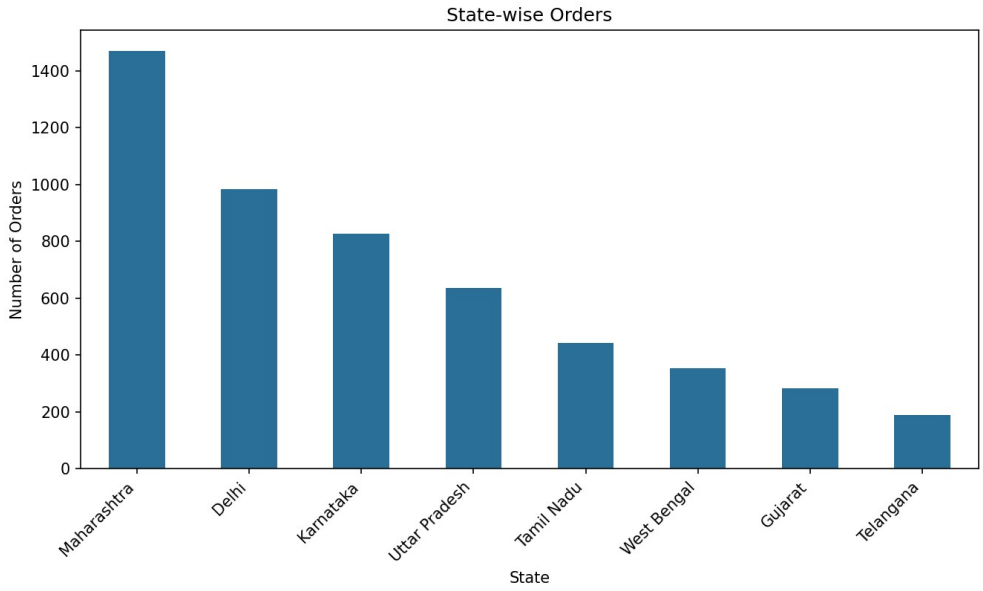


Figure 5A: Orders by state.

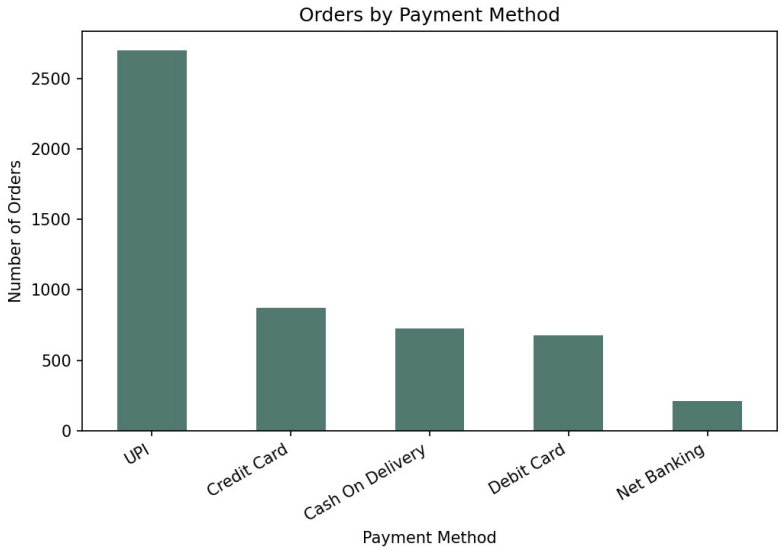


Figure 5B: Orders by payment method.

The figures show **state-wise order volume** and **payment method distribution** recorded during the 30-day period. The X-axis in Figure 5A represents the 8 states in the dataset, while the Y-axis represents total orders generated by customers from each state. Figure 5B compares the five payment methods used across all orders.

Observation

The data shows that **Maharashtra** generated the highest number of orders (**1,469**), making it the largest customer segment in the dataset. **Delhi (984)** ranked second, followed by **Karnataka (826)** and **Uttar Pradesh (636)**, while **Tamil Nadu, West Bengal, Gujarat, and Telangana** contributed smaller shares. Nationally, **UPI** was the most preferred payment method at **52.1% of all orders**, followed by **Credit Card (16.8%)**, **Cash on Delivery (14.0%)**, **Debit Card (13.0%)**, and **Net Banking (4.0%)**. In **Maharashtra** specifically, UPI usage was even higher at **55.9%**, reflecting strong digital-payment adoption in major metro markets.

Business Insight

Flipkart should continue strengthening UPI as the default checkout option and prioritize region-specific promotions in high-volume states such as Maharashtra, Delhi, and Karnataka. Since UPI already dominates in these markets, offering **UPI-specific cashback or instant-discount campaigns** can further increase conversion and reduce payment-failure drop-

off compared with net banking. Expanding fulfilment capacity and warehousing in top states can also reduce delivery times, while targeted marketing in lower-volume states (Telangana, Gujarat) could unlock untapped demand.

Summary of Key Metrics

Objective	Key Metric
Most sold product	boAt Airdopes 141 (1,132 units)
Total orders analyzed (clean)	5,179
Total revenue	₹2,11,68,070
Average order value	₹4,087
Average daily orders	~172.6 / day
Peak order day	10 September 2026 (327 orders)
Peak order hour	8 PM–9 PM IST
Highest rated category	Books (4.56)
Highest return-rate category	Fashion (11.74%)
Largest customer base (state)	Maharashtra (1,469 orders)
Most preferred payment method	UPI (52.1%)
Overall return rate	6.64%
Overall delivered rate	91.2%
Top device used for ordering	Mobile App (77.6%)

How This Would Work at Real Scale

The dataset used in this report is synthetic and created for academic purposes, but the same descriptive analytics approach can be applied to Flipkart's real e-commerce platform. In a production environment, millions of customers generate order data every day through browsing, cart additions, purchases, ratings, returns, and payment transactions. Each interaction — viewing a product, applying a coupon, selecting a payment method, completing checkout, or requesting a return — would be recorded and analyzed.

At real scale, this data would be collected across devices and regions and processed through cloud-based analytics pipelines. The platform could automatically track the most ordered products, daily order trends, peak order hours, category performance, and regional customer preferences. These insights would help Flipkart optimize homepage recommendations, sale-event timing, warehouse placement, delivery logistics, and payment-gateway partnerships.

For example, if customers in Maharashtra consistently prefer **UPI** and show the highest engagement between **8 PM and 9 PM**, the platform could prioritize UPI incentives, schedule flash sales during peak hours, and target promotions to that region. Similarly, category-level return rates and ratings would help identify high-performing product lines and flag categories needing better quality control or catalog information.

Overall, descriptive analytics at real scale would provide real-time insight into customer behavior, product performance, and regional demand, enabling Flipkart to make faster, data-driven business decisions.

Layer / Purpose / Typical Tools

Layer	Purpose	Typical Tools
Data generation	Create a synthetic Flipkart order dataset for analysis	Python, Microsoft Excel
Data processing	Clean, organize, and prepare the dataset for descriptive analysis	Python (Pandas), Jupyter Notebook (Anaconda)
Storage	Store the dataset in CSV/Excel format for analysis and visualization	Microsoft Excel (.xlsx), CSV
Data visualization	Create charts and identify trends and customer patterns	Matplotlib, Jupyter Notebook (Anaconda)
Report preparation	Interpret results, prepare insights, and format the final analytics report	Microsoft Word, Claude

Next Steps to Move From Descriptive → Predictive

The descriptive analytics performed in this report provides a strong foundation for future predictive analytics. The next step would be to use historical order data to forecast future customer behavior, product demand, and engagement trends. Machine learning models could predict which products are likely to trend, which customers may churn, and which customers are most likely to upgrade to **Flipkart Plus**. Predictive models could also estimate peak order demand, improve product recommendations, optimize regional payment incentives, and support state-level inventory planning.

The predictive analytics stage can focus on the following key areas:

- Predict future product demand using historical order trends.
- Forecast peak order hours for better server and logistics planning.
- Identify customers at risk of churn to improve retention.
- Personalize product recommendations based on order history and category preferences.
- Optimize UPI incentives and regional marketing spend for high-volume states.

Conclusion

This descriptive analytics study successfully analyzed 30 days of synthetic Flipkart order data to understand product popularity, daily order trends, peak order hours, category ratings, return rates, and state-wise customer engagement. The analysis showed that **boAt Airdopes 141** was the most sold product, **8 PM–9 PM** was the peak ordering period, **Books** received the highest average customer rating, and **Fashion** recorded the highest return rate. The findings also highlighted that **Maharashtra** was the largest customer segment, with **UPI** being the most preferred payment method.

Overall, the study demonstrates how descriptive analytics can be used to identify customer behavior patterns, improve merchandising and category strategy, support regional and payment-method decisions, and provide a strong foundation for future predictive analytics and recommendation systems.

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Dataset Citation

Synthetic Flipkart Sales Dataset (30 Days), generated for academic descriptive analytics purposes using simulated order behavior, Indian state-level preferences, and realistic e-commerce engagement patterns.

Appendix

Appendix Item	Description
Appendix A	Sample of the synthetic Flipkart order dataset (30 days)
Appendix B	Python code used for data generation, cleaning, and preparation
Appendix C	Python code used for KPI calculations
Appendix D	Python code used for data visualization (Matplotlib)
Appendix E	Category-wise average rating table
Appendix F	Category-wise return rate table
Appendix G	State-wise order count table
Appendix H	Hour-wise order count table
Appendix I	Daily order trend data
Appendix J	Additional charts and supporting analysis tables